

SIMATS ENGINEERING

CO3- AT2 - Design Task - Medium

COURSE NAME: Advance Wireless Communication Systems /

5G / 6G Communication Systems

COURSE CODE: ECA56

COURSE OUTCOME COVERED

CO: 3 - Design spectrum access strategies, waveform generation methods, and multiple access techniques for efficient wireless transmission systems. (BL6)

Assessment Tool Weightage: 35%

Short description about assessment: Students design a system / model based on given requirements to assess creativity and application skills.

Dr.

QUESTIONS: 1 (1 X 100 = 100)

Total Marks: 100 Duration: 3hrs

1. Design a system architecture for Multi-access Edge Computing to reduce latency in smart traffic applications. (BL6)
2. Develop a cloud-edge communication model for wireless healthcare monitoring. (BL6)
3. Design a functional architecture for real-time edge data processing in industrial automation. (BL6)
4. Propose a cloud-based wireless framework for smart surveillance systems. (BL6)
5. Design an edge AI workflow for intelligent wireless data filtering. (BL6)
6. Develop a task offloading strategy between edge and cloud in 5G applications. (BL6)
7. Design a low-latency communication architecture for edge-assisted smart city services. (BL6)
8. Propose a scalable edge-cloud framework for wireless sensor networks. (BL6)
9. Design a Cloud Radio Access Network model for dense urban deployment. (BL6)
10. Develop a functional split model between centralized and distributed radio units. (BL6)
11. Design a virtualized wireless access framework for next-generation communication. (BL6)
12. Propose a data flow model for cloud-based radio resource allocation. (BL6)
13. Design a centralized traffic control model using Cloud RAN principles. (BL6)
14. Develop a network management architecture for cloud-assisted wireless systems. (BL6)
15. Design a wireless service framework integrating virtual network functions.

- (BL6)
16. Propose an optimized fronthaul communication model. (BL6)
 17. Design a satellite-assisted communication model using Low Earth Orbit satellite for remote wireless coverage. (BL6)
 18. Develop a comparative architecture for LEO, MEO, and GEO backhaul systems. (BL6)
 19. Design a spectrum sharing model between LTE and 5G users. (BL6)
 20. Propose a frequency allocation strategy for mixed wireless services. (BL6)
 21. Design a dynamic spectrum access framework for urban wireless systems. (BL6)
 22. Develop a communication architecture combining terrestrial and satellite networks. (BL6)
 23. Design a spectrum efficiency improvement model for dense communication zones. (BL6)
 24. Propose a channel reuse plan for multi-user wireless communication. (BL6)
 25. Design an NB-IoT architecture for smart utility monitoring. (BL6)
 26. Develop an LTE-M communication workflow for wearable monitoring systems. (BL6)
 27. Design a low-power communication model for battery-operated IoT devices. (BL6)
 28. Propose a device communication plan for long-life sensor networks. (BL6)
 29. Design a 5G-enabled IoT framework for campus automation. (BL6)
 30. Develop an energy-efficient wireless communication strategy for sensor nodes. (BL6)
 31. Design an IoT communication flow for periodic environmental sensing. (BL6)
 32. Propose a scalable wireless architecture for smart connected devices. (BL6)
 33. Design a smart city communication framework for traffic monitoring and surveillance. (BL6)
 34. Develop an energy monitoring communication model for public infrastructure. (BL6)
 35. Design a smart factory information flow for predictive maintenance. (BL6)
 36. Propose a robotic automation communication framework for industrial control. (BL6)
 37. Design a remote patient monitoring communication architecture. (BL6)
 38. Develop a wireless communication workflow for AI-assisted healthcare diagnosis. (BL6)
 39. Design a smart irrigation communication model for crop monitoring. (BL6)
 40. Propose a livestock tracking communication architecture using wireless sensors. (BL6)

Code of Conduct:

I Deepshikha S (192412090) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Question 3: Real-Time Edge Data Processing in Industrial Automation

1. Understanding of Design

The proposed architecture is a real-time edge data processing framework for industrial automation. The design places edge computing nodes close to industrial machines so that sensor data can be collected, processed, analysed, and acted upon locally with low delay. Industrial sensors such as temperature, vibration, pressure, current, and machine-status sensors continuously generate data. Wireless connectivity, including a 5G industrial network, carries the sensor information to an edge gateway or Multi-access Edge Computing (MEC) node. The edge node performs filtering, feature extraction, threshold checking, anomaly detection, and local decision-making. Only important events, summaries, or alarms are forwarded to the cloud for long-term storage, advanced analytics, and centralized monitoring.

The main design objectives are low end-to-end latency, reliable industrial communication, reduced backhaul traffic, fast fault detection, scalable processing, and continuous machine monitoring. The architecture is intended to support real-time industrial functions such as predictive maintenance, machine-condition monitoring, process supervision, and automated alerts.

The major constraints are wireless channel congestion, processing limitations at edge nodes, large volumes of sensor data, strict latency requirements, reliability requirements, cybersecurity, and the need to maintain operation when the cloud connection is temporarily unavailable.

2. Proposed Functional Architecture

The functional data flow of the proposed system is:

Industrial Sensors → Wireless/5G Access Network → Edge Gateway/MEC Node → Real-Time Processing → Local Control/Actuation + Event Forwarding → Cloud/Control Centre

At the sensing layer, sensors attached to machines acquire physical parameters. The wireless access layer transports these measurements to the nearby edge node. The

edge node acts as the main real-time processing layer. It first buffers and filters incoming data, then performs feature extraction and anomaly detection. If a dangerous condition is detected, a local control or alarm action is initiated immediately. Normal or non-critical information is compressed into summaries and sent to the cloud. The cloud provides historical storage, dashboards, fleet-level analysis, model training, and centralized supervision.

3. Functional Blocks

Industrial Sensor Layer: Collects temperature, vibration, pressure, current, speed, and machine-status measurements.

5G / Wireless Connectivity: Provides wireless transport between industrial devices and the edge gateway, supporting low-latency communication.

Edge Gateway / MEC Node: Receives sensor streams and provides local computing resources close to the machines.

Real-Time Processing Layer: Performs filtering, feature extraction, threshold analysis, anomaly detection, and event classification.

Local Control and Actuation: Generates immediate alarms or control commands when critical conditions are detected.

Cloud / Control Centre: Stores historical data and supports dashboards, advanced analytics, centralized monitoring, and model training.

Security and Management: Provides authentication, encryption, access control, device management, and monitoring across the architecture.

4. Evaluation of Design

The edge-assisted architecture improves industrial automation because time-critical processing is performed near the source of the data. A major strength is reduced latency: an edge node can identify a machine fault and generate an alarm without waiting for a remote cloud server. Another strength is reduced wireless backhaul usage because raw sensor streams do not all need to be continuously transmitted to the cloud. The design also improves resilience because local decisions can continue during a temporary cloud or backhaul interruption.

The architecture is scalable because additional machines can be connected to edge gateways and additional edge resources can be deployed when processing demand increases. It also supports centralized cloud analytics for long-term trends and predictive maintenance.

However, the design has limitations. Edge nodes require computing and storage resources, which increases infrastructure cost. If an edge node fails, the machines connected to it may lose local processing capability unless redundancy is provided. Wireless interference or congestion can also affect communication reliability. In addition, industrial edge nodes must be protected against unauthorized access and cyberattacks.

5. Design Justification

Edge processing is selected because industrial automation contains time-sensitive events for which cloud-only processing can introduce unnecessary communication delay. Processing sensor data locally allows rapid anomaly detection and immediate control. The use of a 5G wireless access layer is justified by the need for high-capacity connectivity, low latency, and support for a large number of connected industrial devices. The cloud remains important for functions that do not require immediate response, such as historical data storage, large-scale analytics, centralized dashboards, and machine-learning model training.

The hybrid edge-cloud approach therefore divides computation according to timing requirements: urgent decisions are handled at the edge, while computationally intensive or long-term analytical tasks are handled in the cloud. This improves the balance between latency, bandwidth utilization, scalability, and centralized intelligence.

6. Suggestions for Improvement

- Deploy redundant edge nodes so that another node can take over if the primary edge node fails.
- Use adaptive data filtering so that the amount of information forwarded to the cloud changes according to machine condition and network load.
- Apply dynamic resource allocation at the edge so CPU, memory, and processing capacity are assigned according to real-time workload.
- Use secure authentication, encryption, access control, and continuous monitoring for industrial devices and edge servers.
- Use network slicing or dedicated wireless resources for critical industrial traffic to reduce interference and congestion.
- Maintain local buffering so that important sensor information is not lost during temporary cloud or backhaul failures.
- Use AI-assisted anomaly detection at the edge for faster identification of abnormal machine behaviour.

7. Communication and Data Flow

The sensor continuously samples machine parameters and sends measurements through the wireless/5G access network. The edge gateway receives the data and performs preprocessing such as filtering and normalization. The real-time analytics module calculates a representative condition value and compares it with an acceptable threshold. Normal data is summarized and periodically synchronized with the cloud. If an abnormal condition is detected, the edge node immediately generates a local alert and forwards the event information to the cloud and control

centre.

The simplified flow is:

Sensor Data → 5G/Wireless Link → Edge Buffer → Filtering → Feature Extraction → Anomaly Detection → Local Action / Alarm → Cloud Synchronization.

8. MATLAB Simulation

The following MATLAB simulation models the effect of edge processing on data traffic and alert latency. It compares a cloud-only architecture with an edge-assisted architecture. The simulation assumes 100 industrial sensor streams, an average data rate of 1 Mbps per stream, and a 300 Mbps wireless/backhaul capacity. In the edge-assisted case, 85% of redundant data is filtered locally. The model also compares cloud and edge processing delays.

```
clc;
clear;
close all;

% Real-Time Edge Data Processing in Industrial Automation
% Cloud-only vs Edge-assisted architecture

% System parameters
num_devices = 100;          % Number of industrial sensor streams
data_rate_per_device = 1;   % Mbps per device
network_capacity = 300;     % Mbps

% Latency parameters (ms)
t_sensor = 2;               % Sensor acquisition delay
t_wireless_cloud = 25;      % Wireless + backhaul delay to cloud
t_cloud_processing = 60;    % Cloud processing delay

t_wireless_edge = 3;        % Delay to nearby edge node
t_edge_processing = 15;     % Edge filtering and anomaly detection

% Edge filtering assumption
filtering_efficiency = 0.85; % 85% of redundant data removed

% Total generated traffic
total_data_rate = num_devices * data_rate_per_device;

% ----- Cloud-only architecture -----
cloud_bandwidth = total_data_rate;
cloud_latency = t_sensor + t_wireless_cloud + t_cloud_processing;

% ----- Edge-assisted architecture -----
```

```

edge_bandwidth = total_data_rate * (1 - filtering_efficiency);
edge_local_latency = t_sensor + t_wireless_edge + t_edge_processing;

% Data forwarded to cloud after edge filtering
edge_cloud_latency = edge_local_latency + ...
    t_wireless_cloud + t_cloud_processing;

fprintf('--- Industrial Edge Processing Simulation ---\n\n');

fprintf('Number of devices: %d\n', num_devices);
fprintf('Total generated data rate: %.2f Mbps\n\n', total_data_rate);

fprintf('--- Cloud-Only Architecture ---\n');
fprintf('Backhaul bandwidth required: %.2f Mbps\n', cloud_bandwidth);
fprintf('End-to-end processing latency: %.2f ms\n', cloud_latency);

if cloud_bandwidth > network_capacity
    fprintf('STATUS: CONGESTION - capacity exceeded by %.2f Mbps\n\n', ...
        cloud_bandwidth - network_capacity);
else
    fprintf('STATUS: Bandwidth adequate\n\n');
end

fprintf('--- Edge-Assisted Architecture ---\n');
fprintf('Bandwidth after edge filtering: %.2f Mbps\n', edge_bandwidth);
fprintf('Local edge response latency: %.2f ms\n', edge_local_latency);
fprintf('Total latency for cloud synchronization: %.2f ms\n', ...
    edge_cloud_latency);

if edge_bandwidth > network_capacity
    fprintf('STATUS: CONGESTION\n\n');
else
    fprintf('STATUS: Bandwidth adequate\n\n');
end

% ----- Comparison plot -----
latency_values = [cloud_latency edge_local_latency];
bandwidth_values = [cloud_bandwidth edge_bandwidth];

figure;
bar(latency_values);
set(gca, 'XTickLabel', {'Cloud Only', 'Edge Assisted'});
ylabel('Latency (ms)');
title('Latency Comparison');

```

```
grid on;
```

```
figure;  
bar(bandwidth_values);  
set(gca, 'XTickLabel', {'Cloud Only','Edge Assisted'});  
ylabel('Bandwidth (Mbps)');  
title('Bandwidth Requirement Comparison');  
grid on;
```

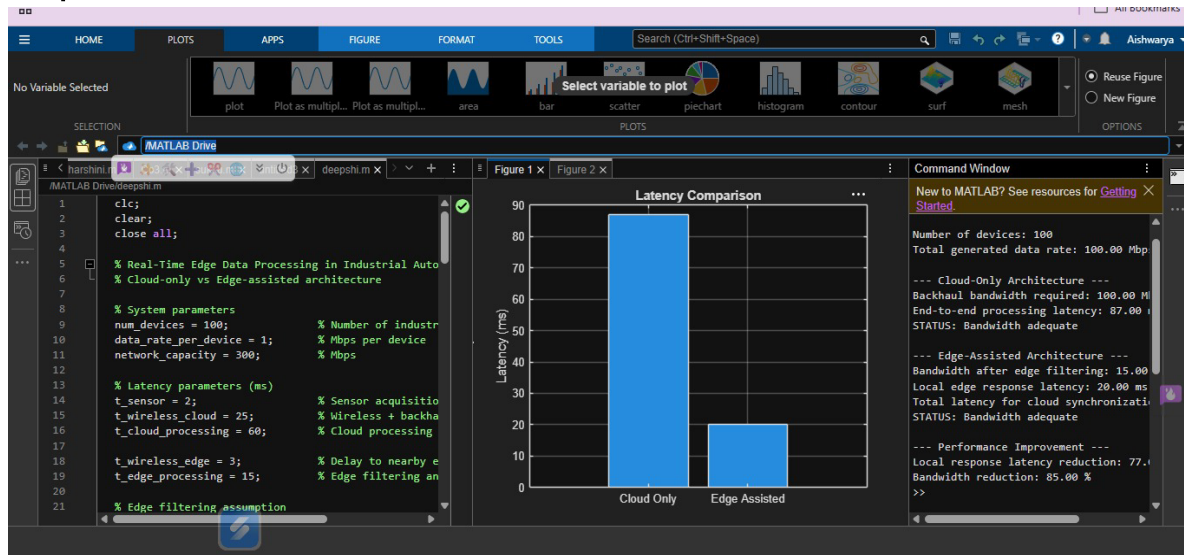
```
% Percentage improvements
```

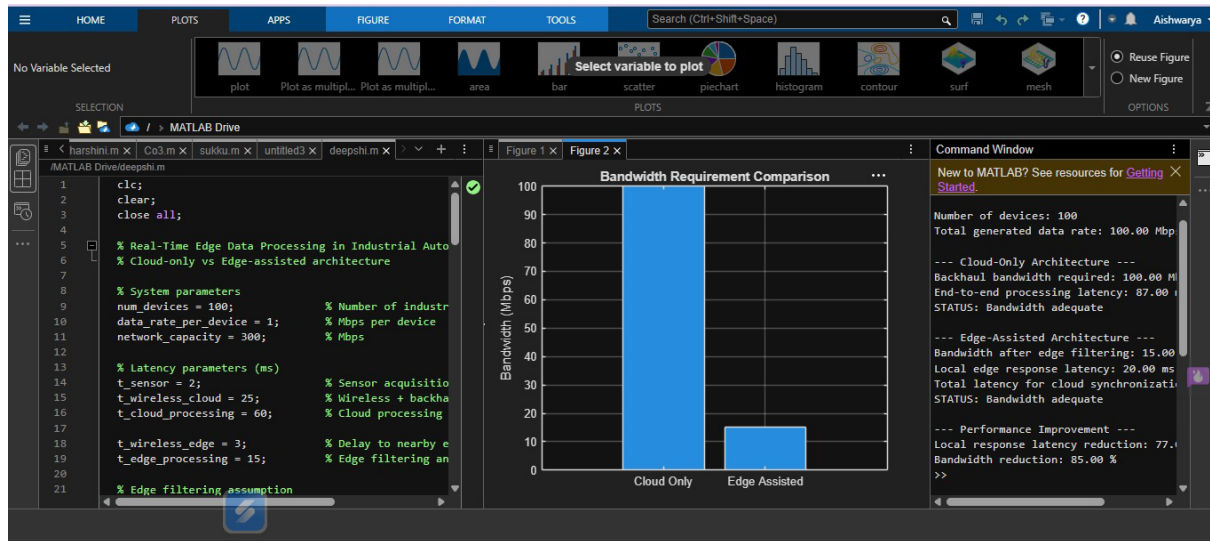
```
latency_reduction = ((cloud_latency - edge_local_latency) / ...  
    cloud_latency) * 100;
```

```
bandwidth_reduction = ((cloud_bandwidth - edge_bandwidth) / ...  
    cloud_bandwidth) * 100;
```

```
fprintf('--- Performance Improvement ---\n');  
fprintf('Local response latency reduction: %.2f %%\n', latency_reduction);  
fprintf('Bandwidth reduction: %.2f %%\n', bandwidth_reduction);
```

Output:





9. Expected Simulation Interpretation

With 100 devices generating 1 Mbps each, the cloud-only architecture produces 100 Mbps of traffic. The edge-assisted architecture filters 85% of redundant information, reducing the forwarded traffic to 15 Mbps. Therefore, the wireless/backhaul load is substantially reduced. The local edge response is also faster because the critical decision is made near the industrial machines instead of waiting for cloud processing.

For the selected parameters, the cloud-only response latency is 87 ms, while the local edge response latency is 20 ms. This corresponds to a substantial reduction in the response time for critical events. The bandwidth requirement falls from 100 Mbps to 15 Mbps, demonstrating the main benefit of local edge filtering. The cloud can still receive selected events and summaries for synchronization and long-term analysis.

10. Conclusion

The proposed functional architecture combines industrial sensors, 5G/wireless connectivity, edge computing, real-time analytics, local control, and cloud services. The design places time-critical processing at the edge to reduce latency and unnecessary data transmission, while the cloud provides centralized storage and advanced analytics. The MATLAB simulation demonstrates that edge filtering can significantly reduce bandwidth usage and improve response time. With redundancy, secure communication, adaptive resource allocation, and intelligent anomaly detection, the architecture can provide a practical and scalable foundation for real-time industrial automation.

RUBRICS FOR CALCULATION:

Design Task					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Design	20%	Demonstrates deep understanding of design objectives, constraints, and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Lacks understanding of the design context
Evaluation of Design	25%	Provides in-depth critique identifying strengths, weaknesses, and design gaps	Identifies key issues with reasonable analysis	Limited critique; mostly descriptive feedback	No meaningful critique provided
Justification	20%	Supports critique with strong reasoning, evidence, and relevant principles	Provides justification with some supporting evidence	Weak or unclear justification	No justification for feedback
Suggestions for Improvement	20%	Offers innovative, practical, and well-justified improvement suggestions	Provides useful suggestions with minor gaps	Suggestions are limited or partially relevant	No constructive suggestions provided
Communication	15%	Communicates feedback clearly, professionally, and engages actively in discussion	Communicates well with minor issues	Communication lacks clarity or engagement	Poor communication and minimal participation